

Ege Sarp Dengizmen

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SUMMARY

Mechanical Engineer with interdisciplinary capability across electronics, embedded systems, and computer science. High Honour graduate from Middle East Technical University with two years of part-time structural design experience at Turkish Aerospace Industries. My technical focus lies in mechatronic system design, feedback control, real-time embedded implementation, and CAD-driven hardware development. I have hands-on experience integrating sensors, actuators, and microcontrollers (e.g., Arduino, ESP32) into functional prototypes. Skilled in FDM-based rapid prototyping and iterative testing. I am particularly motivated by applied mechatronics, hardware-centric R&D, vision based measurements and the development of robust, physically realizable control solutions.

EXPERIENCE

Structural Design Engineer (Part Time)

Turkish Aerospace Industries

April 2022 - June 2024, Ankara

- Performed research and presentations on implementations of various design solutions for the Turkish Fighter X.
- Designed structural parts for the horizontal and vertical tail of the Turkish Fighter X, using Catia V6.
- Automated various repetitive procedures performed during design using Catia V6 VBA macros to aid my team.

Manufacturing Intern

Titan Makina Ltd. Şti.

August 2021 - September 2021, Ankara

- Worked in both production and assembly departments, coordinating workflows and ensuring effective inter-departmental communication.
- Created and distributed work orders, monitored progress, and ensured parts were produced to specification and delivered on schedule.
- Monitored inventory levels, checked stock status, and generated purchase requests for required components.
- Developed hands-on knowledge of TIG and MIG welding, as well as additional manufacturing and fabrication techniques.

EDUCATION

Mechanical Engineering Master

Politecnico di Milano • Milano, MI • September 2024 - Present

- Mechatronics and Robotics Track.

Mechanical Engineering Bachelor

Middle East Technical University • Ankara • September 2018 - June 2024 • 3.6

High School Education

The Koç School • Istanbul • September 2014 - June 2018

- Diploma average: 85

PROJECT

METU ROMER - MIDIBOT

Middle East Technical University • github.com/Bayrakt4rdem/MIDI_ROBOT • January 2023 - June 2023

- Structuring and coding of the Python API for the robot.
- Design and optimization of mechanical components of the robot for 3D printing.
- Docker container setup, allowing for fast and easy setup of a Raspberry Pi for operation.
- Coding of the micro-controller for sensor integration and stepper control.
- Setup of the ROS2 workspace with launch files and some essential nodes for robot control through twist messages, transforms, state publishing, serial communication, robot URDF and Rviz.

Design and Construction of a Small-Scale Vertical Wind Turbine for Urban Application

Middle East Technical University • September 2023 - January 2024

- Calculations of energy demand and expected energy output.
- Design and preliminary calculations for sizing of mechanical components.
- Performed finite element analyses to validate mechanical component sizings for safe operation.
- Selection and integration of electronic components to meet demand and storage capacity.
- Circuit design for emergency electrical braking based on shaft RPM.
- Experimental testing of the generator to validate energy output.

Optimization and Application of a Tuned Mass Dampener (TMD)

Middle East Technical University • drive.google.com/drive/folders/1zP-n07zpuOsah92MvKS0yBe-DmnwEBjq?usp=drive_link • July 2023 - August 2023

- Free vibration response was measured using an IMU, and the data was passed through a low pass filter to smooth out noise in the measurements.
 - Logarithmic decrement method was used to determine vibrational parameters from experimental data.
 - Optimizations were performed to identify the TMD vibrational properties which were then implemented into the prototype followed by testing.
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INVOLVEMENT

Volunteer | Board of Management

Middle East Technical University • METU Animal Friends Club • September 2020 - July 2025

- Organizing a team of volunteers to ensure the safety and well being of the animals on our campus on a daily basis.
- Working with a team of designers to produce and sell products that create considerable revenue.
- Coordinating and organizing weekly, charity based yoga events to generate revenue for the club.

Athlete

Middle East Technical University • METU Snowboard Team • December 2021 - March 2024

- Competed in several nationwide and university boardercross tournaments.

Volunteer

Middle East Technical University • METU Robotics Club • September 2018 - January 2021

- Education Team: Teaching of newcomers about Arduino, Python, and C programming.
 - Drone Team: Working on image recognition and object detection with OpenCV.
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SKILLS

CAD: Siemens NX, CATIA V6, Fusion 360, Onshape, Catia-VBA

Programming: Python, Matlab/Simulink, Docker, ROS2, C

Office Products: Word, Excel, PowerPoint

Creative Products: Photoshop, Premiere Pro

Microcontrollers: Arduino, Raspberry Pi Pico, ESP32

Languages: Turkish, English, German
